Breaking the Silence: Large Language Models for Culturally-Sensitive Menopause Education

AI Ethics, Cultural Sensitivity & Health Informatics

Student Team Presentation

Computer Science Department

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Outline

Assignment Summary

Core Objective

Design and develop a culturally-sensitive AI system for menopause education, addressing global health education gaps through personalized Large Language Model applications.

Key Topics:

- Large Language Models
- AI Ethics & Safety
- Cultural AI
- Health Informatics
- Bias Detection
- Human-Computer Interaction

Target Audience:

- High School Students (Grades 9-12)
- CS1 College Students

Timeline:

- Phase 1: 2-3 weeks (Research/Design)
- Phase 2: 3-4 weeks (Implementation)
- **Total: 5-7 weeks**

Project Strengths & Challenges

Strengths

- Addresses real-world health equity issues
- Combines technical skills with social impact
- Teaches AI ethics through practical application
- Culturally inclusive and globally relevant
- Integrates multiple AI concepts

Challenges

- Requires sensitive handling of health topics
- Cultural research requires substantial time
- Technical implementation requires programming experience
- May be challenging without health background

Phase 1: Understanding the Global Crisis

Objective

Understand the global menopause education crisis and design a culturally-sensitive AI solution

Research Areas:

- Global menopause education gaps
- Cultural perspectives on menopause
- Current AI applications in health
- Ethical considerations
- Technical feasibility assessment

Deliverables:

- Literature review
- Cultural analysis
- System design proposal
- Ethics framework

Cultural Assessment Framework

Core Concept

Different cultures view menopause differently - from "liberation" to "medical condition" to neutral transition

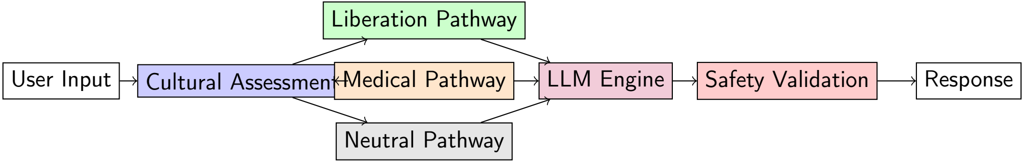
Liberation View	Medical View	Neutral View
Freedom & wisdom	Condition requiring treatment	Normal life stage
Empowerment focus	Symptom management	Balanced information
Traditional communities	Urban/modern lifestyle	Mixed influences
Community guidance	Professional medical advice	Independent learning

Key Insight: AI system must adapt content based on user's cultural perspective

Phase 2: System Architecture

Objective

Build a working prototype of the AI-powered menopause education system



Content Routing System

Adaptive Content Delivery

System routes users to appropriate content pathways based on cultural assessment

Listing 1: Content Router Implementation

```
class ContentRouter:
    def __init__(self):
        self.pathways = {
            "liberation": LiberationPathway(),
            "medical": MedicalPathway(),
            "neutral": NeutralPathway()
        }

    def route_user(self, assessment_results):
        """Route user to appropriate content pathway"""
        score = self.calculate_pathway_score(assessment_results)
        return self.pathways[score["dominant_pathway"]]
```

Safety Validation Layer

Critical Component

Multi-layered safety validation to prevent misinformation and ensure cultural sensitivity

Listing 2: Safety Validator

```python
class SafetyValidator:
    def validate_response(self, response, user_context):
        """Multi-layer safety validation"""
        medical_check = self.verify_medical_accuracy(response)
        bias_check = self.detect_cultural_bias(response, user_context)
        confidence_check = self.assess_confidence(response)
        content_check = self.validate_content_appropriateness(response)

        return {
            "approved": all([medical_check, bias_check,
                             confidence_check, content_check]),
            "requires_human_review": not medical_check,
            "safety_flags": self.get_safety_flags(response)
        }
```

Bias Detection Framework

Identifying Cultural Bias

System monitors for various types of bias in AI responses

Bias Type	Description	Weight
Western Centrism	Imposing universal standards	0.8
Medical Bias	Pathologizing natural experiences	0.9
Age Bias	Negative framing of aging	0.7
Gender Bias	Stereotypical assumptions	0.9
Socioeconomic Bias	Assuming expensive treatments	0.6

Key Features:

- Real-time bias scoring
- Cultural appropriateness assessment
- Automated bias reduction recommendations

Assessment Rubric

Component	Excellent (90-100%)	Good (80-89%)	Satisfactory (70-79%)	Needs Improvement (<70%)	Weight
Research Quality	Comprehensive research with diverse sources	Good research with credible sources	Adequate research	Limited research	20%
Technical	Fully functional system	Mostly functional with minor bugs	Basic functionality	Major functionality missing	25%
Cultural Sensitivity	Exceptional cultural awareness	Good cultural consideration	Basic awareness	Little cultural consideration	20%
AI Ethics & Safety	Comprehensive safety mechanisms	Good safety considerations	Basic safety measures	Inadequate safety	20%
Presentation	Clear, engaging presentation	Good presentation	Adequate presentation	Poor communication	15%

Project Variants

Simplified Version

For Beginners:

- Focus only on Phase 1 research and design
- No technical implementation required
- Emphasis on cultural understanding
- Paper prototype development

Cross-Curricular

Interdisciplinary Approach:

- Partner with health/biology classes
- Collaborate with social studies
- Include psychology perspectives
- Medical professional interviews

Advanced Version

For Experienced Students:

- Implement machine learning models for bias detection
- Advanced NLP techniques

Global Perspective

International Scope:

- Compare multiple cultural contexts
- Beyond India - explore other cultures
- Cross-cultural validation

Prerequisites & Dependencies

High School Level

Required:

- Basic understanding of AI concepts
- Internet research skills
- Presentation abilities
- Cultural sensitivity awareness

Helpful:

- Basic programming exposure
- Health science background

CS1 College Level

Required:

- Python programming fundamentals
- Basic web technologies (HTML/CSS)
- Introduction to APIs
- Data structures knowledge

Helpful:

- Machine learning basics
- Database concepts
- UI/UX design principles

Key Learning Outcomes

Students Will Learn:

- AI ethics in healthcare
- Cultural sensitivity in technology
- Bias detection and mitigation
- LLM applications and limitations
- Health informatics principles
- Human-computer interaction
- Safety-critical system design
- Global health equity issues
- Research and analysis skills
- Technical implementation
- Presentation and communication
- Interdisciplinary collaboration

Breaking the Silence

This project addresses a real-world health crisis affecting over 1 billion women globally while teaching cutting-edge AI concepts through practical, ethical application.

Technology + Ethics + Cultural Sensitivity = Real Impact

Remember

Success depends not just on technical implementation, but on cultural understanding, ethical considerations, and commitment to health equity.

Thank You!

Questions & Discussion

"AI can break the silence, but only with wisdom, ethics, and cultural respect."